

**DRIVER DECISION SUPPORT SYSTEM FOR RAILWAY
SAFETY**

25-26J-134

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

Specializing in Data Science

Department of Computer Science

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DECLARATION

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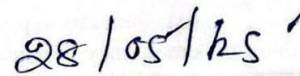
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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



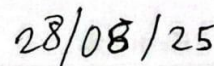
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Date

ABSTRACT

Railway transport remains an essential mode of mobility in Sri Lanka. However, safety concerns persist, with human error identified as a major contributor to train-related accidents. Drivers often operate under monotonous conditions for extended hours, making them vulnerable to fatigue, distraction, or sudden health-related emergencies. While modern signaling systems enhance operational efficiency, they lack mechanisms to assess and respond to the driver's real-time physical or cognitive state, creating a critical safety gap. This research proposes an AI-powered Driver Decision Support System (DDSS) tailored to the Sri Lankan railway system. The system leverages multimodal data, including physiological signals (e.g. heart rate), visual cues (e.g. facial expressions, gaze tracking) and environmental factors (e.g. track obstacles). By utilizing adaptive machine learning models, the DDSS can evolve with new data to detect risks more accurately compared to traditional rule-based methods. In safety-critical scenarios such as extreme fatigue, distraction, or pathological conditions, the system will not only issue alerts but also provide proactive decision support ranging from driver notifications to control center alerts, speed regulation, or automated interventions. To address the scarcity of local datasets, the project will implement a simulator-based data generation pipeline enabling the recreation of realistic operational and emergency conditions. Anticipated outcomes include a functional DDSS prototype for Sri Lanka Railways, a unified safety scoring mechanism. It is expected that the proposed system will reduce accident risks, improve passenger safety, and strengthen public trust in railway transport. Moreover, the findings will contribute to global advancements in intelligent transportation safety and provide a foundation for further research and commercial applications.

Keywords

- Railway Safety
- Driver Decision Support System
- Artificial Intelligence
- Multimodal Data
- Intelligent Transportation

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
DDSS	Driver Decision Support System
DMS	Driver Monitoring System
ECG	Electrocardiography
EEG	Electroencephalography
GNN	Graph Neural Network
ML	Machine Learning
SVM	Support Vector Machine
YOLO	You Only Look Once (Object Detection Model)
Unity	Unity Game Engine
OpenDS	Open Driving Simulator
CARLA	Car Learning to Act (Autonomous Driving Simulator)
ML/DL	Machine Learning / Deep Learning

Table 1: List of abbreviations

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1. INTRODUCTION

In railway operations, the safety and reliability of transportation heavily depend on the vigilance and physical health of train drivers. However, train operators often work alone, over extended periods, and under monotonous conditions, increasing the risk of fatigue, distraction, and even sudden medical emergencies. Traditional rule-based systems and manual observation methods are insufficient in detecting and responding to such issues in real time.

Recent advances in artificial intelligence and sensor technologies offer the opportunity to develop intelligent, adaptive systems capable of continuously assessing a driver's condition and surrounding environment. This research proposes an AI-powered Driver Decision Support System (DDSS) that integrates multimodal data including physiological signals, facial expressions, environmental inputs, and operational metrics to monitor driver states and predict potential safety risks. The system is designed to detect not only fatigue and distraction but also critical pathological states such as loss of consciousness, enabling timely interventions to prevent accidents.

By providing real-time alerts and risk scoring, the DDSS supports proactive decision-making and can be integrated into train cabins or driver training programs. This approach aims to enhance railway safety, reduce human-error-related incidents, and lay the groundwork for real-world deployment, particularly in developing countries like Sri Lanka, where improving transport safety is of paramount importance.

1.1 Background & Literature Survey

The integration of Artificial Intelligence (AI) in transportation safety systems has become increasingly vital in recent years. In particular, Driver Decision Support Systems (DDSS) aim to monitor, assess, and respond to driver behavior in real time to reduce accidents and improve operational safety. These systems have been widely researched in the automotive sector but are still underexplored in railway

applications, despite the high stakes involved in train operations where human error can lead to catastrophic consequences.

In railway systems, drivers often operate alone for extended periods, facing fatigue, distraction, and occasionally undetected medical emergencies. Unlike road-based vehicles, trains have a longer stopping distance, fixed paths, and stricter signaling systems making the margin for driver error extremely narrow. Hence, a proactive DDSS tailored specifically for trains, integrating multimodal data sources and powered by adaptive AI algorithms, is essential for improving safety outcomes.

Ensuring driver safety is a critical component in railway operations, where human error remains a leading cause of accidents. According to a global review of railway incidents by [1], most of fatal accidents in rail transportation involve human factors such as fatigue, inattention, or health-related impairments. In many railway systems, especially in countries like Sri Lanka, trains are often operated by a single driver for extended periods and under monotonous conditions amplifying the risks of cognitive or physiological decline during operation.

The earliest attempts to support train drivers and ensure operational safety relied on rule-based and fuzzy logic systems. These approaches aimed to simulate human decision-making by encoding a set of predefined rules and linguistic variables such as “slightly fatigued” or “moderately alert.” According to [2], fuzzy logic was particularly useful in managing uncertainty and vague conditions such as assessing fatigue levels based on eye blink rate or response delay.

However, while fuzzy logic enabled systems to operate under ambiguity, these traditional methods had major limitations:

- They lacked scalability: new rules had to be manually added for every possible situation.
- They could not adapt to individual driver patterns or changing environments like weather, tunnel conditions, or health anomalies.

- They were reactive, not predictive delaying critical interventions in real emergencies.

With the growing complexity of real-world rail systems, these rule-based architectures were increasingly seen as insufficient.

Recent advancements in AI, deep learning, and embedded systems have led to a paradigm shift in [3] driver monitoring. Multimodal Driver Monitoring Systems (DMS) now combine:

- Visual data (e.g. facial expressions, gaze, head pose),
- Physiological signals (e.g. EEG, ECG, heart rate),
- Environmental conditions (e.g. visibility, obstacles),

For instance, [4], applied convolutional neural networks (CNNs) to analyze eye closure, yawning, and facial micro-expressions to detect fatigue in drivers, achieving good accuracy. Similarly, [5] EEG data processed through machine learning models like SVM and ANN to detect fatigue, demonstrating the potential of bio-signals for safety-critical tasks. More recently, YOLO-based deep learning models have been adapted for real-time train driver drowsiness detection, where YOLOv8 outperformed earlier versions in both accuracy and efficiency [6].

The recent introduction of Graph Neural Networks (GNNs) in DDSS has further improved classification accuracy by capturing complex spatial-temporal relationships between facial and skeletal features. A study by [7] developed a GNN model for classifying driver states including pathological conditions using posture and facial expression data, achieving high accuracy in some scenarios.

However, most of these systems are still:

- Designed for road vehicles, not railway environments.
- Focused only on fatigue or distraction, ignoring critical health issues like stroke or unconsciousness.

- Trained on limited or proprietary datasets, lacking generalization to diverse driver demographics or operational conditions.

Moreover, there is a shortage of publicly available multimodal datasets for railway driver monitoring. Studies by [8] emphasize the importance of synthetic data generation using simulators like Unity, CARLA, and OpenDS for testing DMS models in rare or dangerous situations (e.g. stroke onset or signal failure under fog). Synthetic datasets allow safe and reproducible testing of ML models, particularly in the absence of real-world medical incident data [9].

This research contributes to the field by proposing a simulator-enhanced, AI-driven DDSS for trains, capable of integrating multimodal data in real time. It introduces pathological driver state recognition and a unified safety score system, addressing gaps in both academic research and industry practice.

The proposed system aims to reduce driver-related rail accidents, enhance the trustworthiness of public transport, and provide a benchmark synthetic dataset to support future research. The solution is especially relevant to developing countries like Sri Lanka, where train safety upgrades are needed but constrained by limited real-time monitoring infrastructure.

1.2 Research Gap

Although AI-driven Driver Decision Support Systems (DDSS) have shown promising results in road transportation, their adaptation to railway environments remains minimal. The operational context of trains characterized by fixed tracks, longer braking distances, and strict signaling systems differs significantly from that of road vehicles. However, most existing driver monitoring technologies have been developed for automotive applications without tailoring their design, algorithms, or data processing methods to meet the unique safety and operational demands of railway systems.

Current research in driver monitoring systems predominantly focuses on detecting fatigue and visual distraction. While these factors are important, they represent only part of the safety challenge in railway operations. Other critical conditions such as heart attacks, strokes, or sudden loss of consciousness can occur during prolonged, monotonous train journeys, yet remain underrepresented in existing monitoring frameworks. This narrow scope leaves a significant safety gap, particularly in single-driver railway operations common in countries like Sri Lanka.

Multimodal monitoring approaches have been explored in isolation, using visual cues, physiological signals, or operational data separately. However, there is a lack of fully integrated frameworks that combine these diverse data streams into a unified, real-time safety score. Such an integrated system could enable predictive alerts and proactive interventions, but this approach is rarely implemented in railway contexts.

Traditional rule-based and fuzzy logic methods, though useful for handling ambiguity, lack the adaptability needed for dynamic railway conditions. They cannot learn from individual driver patterns or adjust to environmental variations such as weather, tunnels, or visibility changes. Moreover, these methods tend to be reactive rather than predictive, limiting their ability to prevent incidents before they escalate.

One of the most significant barriers to railway-specific DDSS development is the lack of publicly available multimodal datasets. Existing datasets are either proprietary, focused on road scenarios, or limited to narrow use cases. Rare but critical events, such as a driver experiencing a medical emergency while the train is in motion, are often absent from real-world data due to their infrequency and safety risks during collection.

While synthetic data generation through simulators such as Unity, CARLA, or OpenDS is gaining traction in automotive research, its use in railway safety studies is still underdeveloped. The potential to simulate rare or dangerous railway scenarios such as sudden health deterioration, emergency braking in low visibility, or signal failures remains largely untapped, leaving a gap in training and validating AI models for extreme conditions.

These gaps collectively highlight the urgent need for a railway-specific, AI-powered DDSS that integrates multimodal data, incorporates pathological state detection, utilizes synthetic data for rare events, and applies advanced AI models for adaptive, predictive safety monitoring. Such a system could address current limitations, improve operational safety, and serve as a benchmark for future research in this critical domain.

Aspect	Most Existing Systems	Proposed Railway-Specific DDSS
Domain of Application	Primarily developed for road vehicles (cars, trucks, buses). [1],[3]	Designed specifically for trains with fixed tracks, longer braking distances, and strict signaling systems.
Focus of Monitoring	Primarily focused on basic alertness, drowsiness, and distraction detection.[1],[4],[5]	Includes fatigue, distraction, and pathological conditions
Data Sources	Uses single-modal data (visual OR physiological) in isolation.[5],[6]	Multimodal Real-Time fusion of visual data, physiological signals and environmental conditions into a unified safety score.
Adaptability	Rule-based and fuzzy logic approaches; reactive, static, limited adaptability.[2],[7]	Adaptive AI/ML models (CNN, GNN) that learn driver-specific patterns and provide predictive alerts.
Data Availability	Datasets are proprietary, limited to road scenarios, and lack rare medical/safety cases.[8],[9]	Simulator-based synthetic dataset generation to model rare, high-risk railway scenarios.
Risk Assessment	Reactive, alerts based on a single parameter threshold (e.g. "drowsiness detected").[6],[7]	Dynamic Risk Scoring, Synthesizes multiple risk dimensions into a single, real-time, and interpretable score.
Decision Support	Provides warnings/alerts but no unified decision-making framework.[6],[7]	Introduces a real-time safety scoring system that can issue alerts, notify control center, or trigger automated interventions.

Table 2: Research comparison

1.3 Research Problem

Railway operations remain highly vulnerable to human error, with driver-related factors such as fatigue, distraction, and undetected medical emergencies contributing significantly to accidents. While AI-driven Driver Decision Support Systems (DDSS) have been successfully implemented in road transportation, these solutions are rarely adapted to the specific operational and safety requirements of railways. Existing systems are often limited to detecting fatigue or visual inattention, lack integration of multimodal data sources, and fail to address sudden pathological conditions such as strokes or loss of consciousness.

Furthermore, most current driver monitoring approaches rely on isolated data modalities, traditional rule-based or fuzzy logic frameworks, and datasets that are either proprietary, road-vehicle-oriented, or insufficiently representative of railway-specific conditions. The absence of railway-focused, publicly available multimodal datasets particularly those that include rare but critical events restricts the development, training, and benchmarking of advanced AI models. Synthetic data generation, though widely explored in automotive research, remains underutilized in railway contexts despite its potential to simulate dangerous or uncommon scenarios in a controlled environment.

The problem is compounded by the limited adoption of advanced machine learning architectures, such as Graph Neural Networks (GNNs), which have shown superior performance in spatial-temporal pattern recognition but are seldom applied to railway driver monitoring. Without a predictive, adaptive, and health-aware AI-driven DDSS that addresses these limitations, railway systems especially in developing countries like Sri Lanka will continue to face preventable safety risks and lack the infrastructure needed for real-time driver condition assessment and proactive intervention.

2. OBJECTIVES

The main objective of this research is to address the critical gaps in existing driver decision support systems for railway operations by developing and validating a novel, context-aware, multimodal AI-powered DDSS, specifically tailored to enhance safety and enable proactive intervention in the unique railway environment, especially relevant for regions like Sri Lanka.

2.1 Main Objective

To design, develop, and evaluate a holistic AI-driven Driver Decision Support System (DDSS) that leverages multimodal data fusion, dynamic risk scoring, and advanced machine learning to significantly improve real-time driver state assessment and proactively mitigate safety risks in railway operations

2.2 Specific-Objectives

- **To develop a robust multimodal data integration framework** capable of seamlessly fusing real-time physiological signals (heart rate), visual data (facial expressions, gaze) and environmental conditions (track obstacles) into a unified pipeline for comprehensive driver and situational awareness.
- **To engineer a dynamic and interpretable risk scoring mechanism** that synthesizes inputs from driver state and environmental conditions into a single, adaptive safety score, enabling intelligent alerting and prioritization of interventions based on evolving risk levels.
- **To implement advanced pathological state detection capabilities** within the DDSS, going beyond conventional fatigue/distraction monitoring to accurately identify critical medical conditions such as stroke symptoms, abnormal heart rhythms, or signs of loss of consciousness in real time, crucial for isolated train drivers.

- **To integrate self-learning AI algorithms** (e.g. using Decision Trees, Random Forests, or Neural Networks, potentially including Graph Neural Networks) that enable the DDSS to continuously adapt to individual driver behaviors, environmental variations, and changing operational contexts, thereby improving its predictive accuracy and reliability over time.
- **To establish a simulator-based data generation pipeline** for railway-specific multimodal datasets (utilizing tools like CARLA, Unity, or Open Rails) to address the scarcity of real-world data, creating a synthetic-yet-realistic benchmark dataset for the training, testing, and future research of AI-powered railway DDSS.

3. METHODOLOGY

The methodology for this project is designed to develop a robust, AI-powered Driver Decision Support System (DDSS) for railway operations using simulation-based multimodal data. Each component of the methodology is carefully selected to address the research problem, maximize safety outcomes, and ensure feasibility within the undergraduate project scope.

System Overview

Simulator & Data Generation

Because real-world collection of rare pathological events (stroke, sudden unconsciousness) is unsafe and impractical, all data for this research is generated using high-fidelity railway simulators (Unity/Open Rails). The simulator provides synchronized multimodal datasets, including visual cues (facial expressions, gaze) and physiological signals (heart rate). This ensures comprehensive coverage of both normal and critical scenarios while maintaining safety and reproducibility.

Data Preprocessing Layer

Raw simulated data is cleaned, synchronized, and transformed into features suitable for machine learning. Preprocessing includes noise removal, normalization, feature

extraction, and automatic labeling of driver states and operational events. This step ensures that the data quality supports accurate model training and reflects realistic operational variability.

AI/ML Analysis Layer

The research employs a combination of advanced machine learning models:

- CNNs/LSTMs for visual and temporal patterns in facial expressions and gaze.
- 1D CNNs/GRUs for physiological signal analysis.
- Risk Prediction → GNNs + multimodal fusion combine behavioral + physiological + environmental inputs to predict high-risk situations before they become critical.
- Alert Generation → If the risk score crosses a threshold, the system triggers alerts/warnings to the driver or control center.

Risk Assessment & Decision Support Layer

A dynamic safety scoring mechanism translates multimodal driver state and operational data into actionable insights. High-risk states trigger alerts for the driver or control center, improving operational safety and reducing response time to critical events.

Evaluation & Validation Layer

The system is tested across multiple simulated scenarios to measure accuracy, timeliness, robustness, and generalization. Stress tests, such as sensor dropout or environmental perturbations, are included to validate reliability.

3.1 System Diagram

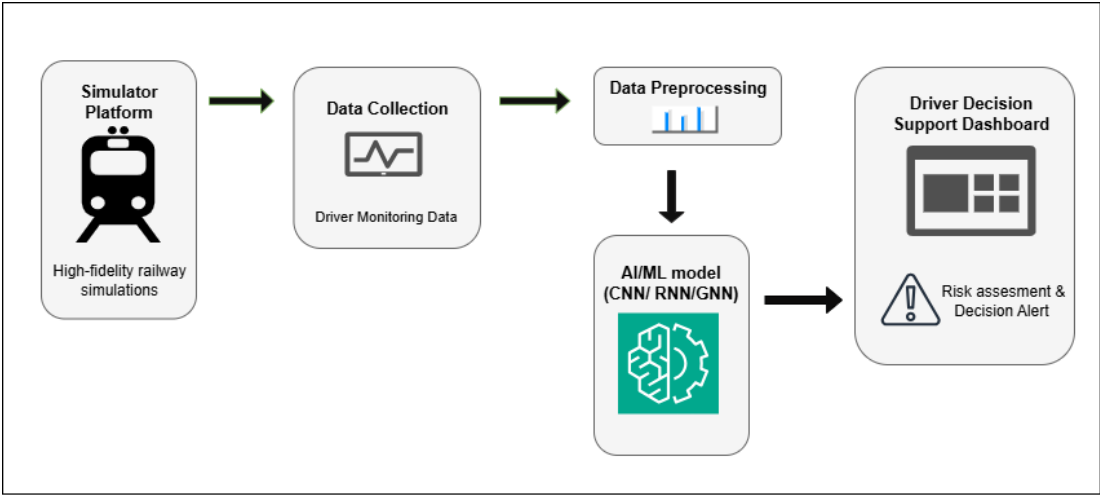


Figure 1: System Diagram

3.2 Work Breakdown Structure

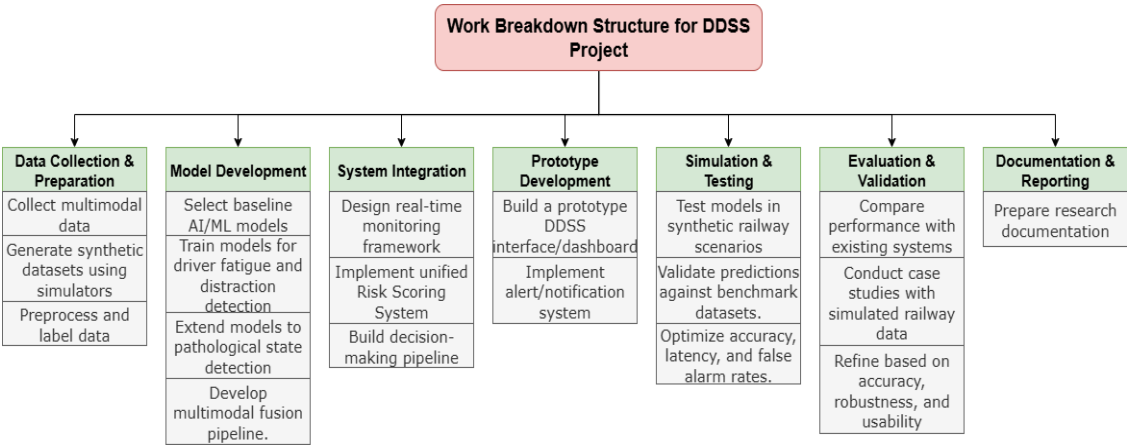


Figure 2: Work Breakdown Structure

3.3 Timeline & Schedule

	2025								2026							
Task	MAY	JUN	JULY	AUG	SEPT	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUN		
Brainstorming Workshop																
Group Registration																
Topic Assessment Form (TAF)																
Project Charter																
Proposal Report (Draft)																
Proposal Presentation																
Proposal Report (Final)																
Implementaion Phrase 1																
Testing 1																
Progress Presentation I																
Implementaion Phrase 2																
Testing 2																
Research Paper																
Progress Presentation II																
Final Report Draft																
Final Presentation & Viva																
Project Website																
Research Logbook																
Final Report																

Figure 3: Gantt chart

3.4 Material & Resources

The successful implementation of this research project requires a combination of hardware, software, and digital resources to ensure both accuracy and efficiency. Since the project leverages simulator-generated data, the reliance on costly real-world experiments is reduced, while still providing realistic conditions for testing and validation.

Hardware

- High-performance workstation with multi-core CPU and at least 16–32 GB RAM for large-scale data handling.
- Dedicated GPU (NVIDIA RTX/GTX series or equivalent) to accelerate deep learning model training and visual recognition tasks.

Software and Tools

- Simulation Platforms
 - Unity 3D / Open Rails Simulator – for generating synthetic, realistic railway driver environments.
- Programming & Model Development
 - Python – primary programming language.
 - TensorFlow / PyTorch – for machine learning and deep learning model training.
 - Scikit-learn – for preprocessing, feature extraction, and evaluation metrics.
- Image & Signal Processing
 - OpenCV – for driver face, eye, and gesture recognition.
 - MATLAB – for signal analysis (heart rate or drowsiness detection).
- Backend Development
 - Flask / FastAPI – for building REST APIs to serve AI models.
- Frontend Development
 - Streamlit – for rapid prototype dashboards (Python-based).
 - React.js + Tailwind CSS – for building professional, interactive user interfaces and operator dashboards.
- Data Management & Storage
 - MongoDB - Best for unstructured/multimodal data (video frame metadata, JSON logs).
 - Cloud storage platforms (Google Drive) – for large-scale dataset backup and sharing.
- Visualization & Reporting
 - Power BI – for analyzing results and presenting trends.
 - MS Word – for documentation and research reporting.
 - MS PowerPoint – for preparing project presentations.

Other Resources

- Graphics & Visual Assets: Avatars, train environment graphics, and simulation design assets to create realistic virtual scenarios.
- Collaboration & Project Management Tools: GitHub for version control, Trello/Jira for task management, and Overleaf for collaborative documentation.
- Reference Materials: Access to academic journals, IEEE/ACM digital libraries, and simulator manuals.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- Real-time Driver Monitoring – Continuously track driver’s physiological signals (heart rate, facial expressions).
- Fatigue & Distraction Detection – Identify early signs of fatigue, drowsiness, or inattention.
- Health Emergency Detection – Detect sudden medical issues such as unconsciousness, irregular heartbeat, or stroke symptoms.
- Risk Scoring System – Generate interpretable safety scores based on combined sensor data.
- Alert & Intervention Mechanism – Provide immediate alerts when risk is detected.
- Data Logging & Reporting – Store session data for analysis, training, and auditing.

4.2 Non-Functional Requirements

- Performance – The system must process and analyze data with less than 2-second delay for real-time alerts.
- Reliability – The system must ensure >95% uptime during operations and minimize false alerts.
- Scalability – It should support monitoring of multiple drivers simultaneously in real-world operations.
- Usability – The user interface (dashboard, simulator integration) must be intuitive, requiring minimal training.
- Security & Privacy – All driver biometric data must be encrypted and comply with data protection regulations.
- Portability – The system should be deployable both in real train cabins and training simulators.
- Maintainability – System modules (ML models, UI) must be modular to allow easy upgrades.
- Accuracy – The AI/ML model must achieve at least 85–90% accuracy in detecting fatigue/distraction states.

4.3 User Requirements

These describe what the end users expect from the system.

- Drivers should be able to receive timely alerts without distraction.
- Operators should be able to view driver performance data remotely.
- Maintenance teams should be able to update and troubleshoot the system easily.
- The interface should be intuitive and require minimal training.
- Reports should be easy to interpret for performance reviews.

5. COMMERCIALIZATION

The proposed Driver Decision Support System (DDSS) demonstrates strong commercialization potential by addressing a critical need in the railway industry, enhancing safety and reducing human-error-related accidents. With its ability to perform real-time monitoring of fatigue, distraction, and pathological states, the system can be marketed as a safety solution for railway operators worldwide.

The creation of a comprehensive multimodal dataset and risk scoring system further adds value, offering opportunities for licensing, collaborations with research institutions, and integration into existing railway monitoring platforms. Beyond deployment in train cabins, the system can also be commercialized in driver training simulators, providing safe, cost-effective, and data-driven training experiences. Its adaptability makes it particularly attractive for developing countries like Sri Lanka, where railway modernization efforts are growing. The system's combination of AI-driven safety assurance and real-world applicability positions it as a scalable entrepreneurial product with potential for partnerships, technology transfer, and global adoption in the transport sector.

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